

GIRRAWEE HIGH SCHOOL

MATHEMATICS

YEAR 12 Extension 1 HSC

Task 1, 2014

Time Allowed: 90 minutes

Name: _____

Instructions:

Examiner: C. McMillan

- Attempt all questions
- Fill in the circle on the multiple choice answer sheet next to the best response for the questions in Part A
- Start each question in Part B on a new page
- All necessary working must be shown
- Marks may be deducted for careless or badly arranged work

PART A (5 marks)

For questions 1-5 circle the best response from the following:

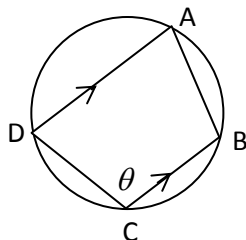
Question 1: The coefficient of x^4 in the expansion $(x - \sqrt{2})^6$ is:

- A) -30 B) 30 C) -60 D) 60

Question 2: The middle term of the expansion $\left(1 + \frac{3x}{4}\right)^6$ is:

- A) $540x^3$ B) $\frac{1215}{4^4}x^4$ C) $\frac{540}{64}x^3$ D) $1215x^4$

Question 3: If $\angle BCD$ is equal to θ then $\angle ABC$ is equal to:

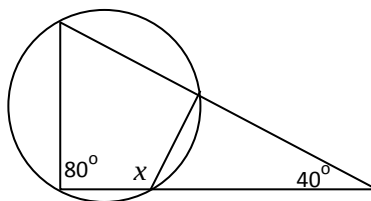


- A) $180 - \theta$ B) θ C) $180 + \theta$ D) 180

Question 4: If $n = k$ is $1 + 2 + 4 + \dots + 2^{k-1} = 2^k - 1$ then $n = k + 1$ is:

- A) $1 + 2 + 4 + \dots + 2^{k-1} + 2^{k+2} = 2^{k-1} - 1$ B) $1 + 2 + 4 + \dots + 2^{k-1} + 2^{k-2} = 2^{k+1} - 1$
C) $1 + 2 + 4 + \dots + 2^{k-1} + 2^k = 2^{k+1} - 1$ D) $1 + 2 + 4 + \dots + 2^{k-1} + 2^k = 2^k - 1$

Question 5: The value of x is:



- A) 120° B) 80° C) 60° D) 40°

PART B

Question 6 (13 marks)

- (a) Prove by mathematical induction that for all positive integers

$$1 + 4 + 7 + \dots + (3n - 2) = \frac{n(3n - 1)}{2}. \quad (5)$$

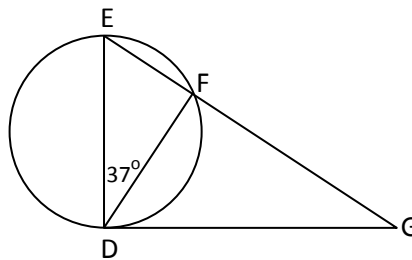
- (b) Using mathematical induction prove that for all positive integers

$$7^{2n} - 3^{3n} \text{ is divisible by } 11. \quad (4)$$

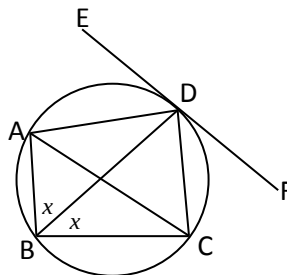
- (c) Show by the principle of mathematical induction that $n! > 2^n$ for $n > 3$. (4)

Question 7 (16 marks) Draw all diagrams in your answer booklet.

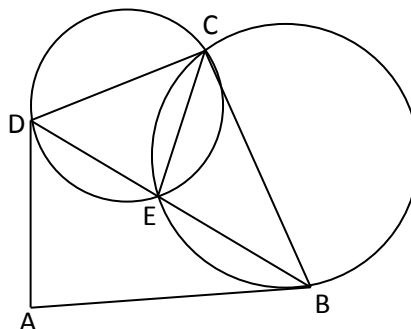
- (a) DE is a diameter of the circle. DG is a tangent. If $\angle EDF = 37^\circ$ calculate the size of $\angle EGD$, giving reasons. (4)



- (b) The diagonal BD of the quadrilateral ABCD, bisects $\angle ABC$. Prove that diagonal AC is parallel to the tangent at D. (6)



- (c) AB and AD are tangents to the circles. DEB is a straight line. Prove that ABCD is a cyclic quadrilateral. (6)



Question 8 (13 marks)

(a) Find the 4th term in the expansion $\left(\frac{m}{2} + 3n\right)^8$. (2)

(b) Find the coefficient of x^5 in the expansion of $(2 - x)^7$. (2)

(c) In the expansion $\left(x - \frac{1}{x}\right)^6 \left(x + \frac{1}{x}\right)^8$, find the coefficient of x^{-8} . (4)

(d) In the expansion $(2 + 3x)^n$, the coefficients of x^3 and x^4 are in the ratio 8:15.

Find n . (5)

Question 9 (15 marks)

(a) Find in the expansion $(5 + 2x)^{12}$:

i) The ratio $\frac{T_{k+1}}{T_k}$, showing all necessary working. (4)

ii) The greatest coefficient. (2)

iii) The greatest term when $x = \frac{1}{2}$. (3)

(b) Find n if the coefficients of the second, third and fourth terms in the expansion of $(1 + x)^n$ are successive terms of an arithmetic series. (6)

Question 10 (14 marks)

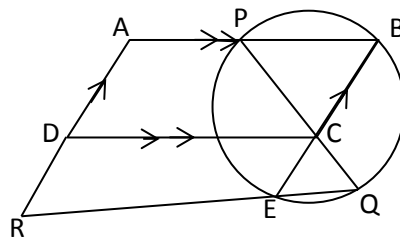
Draw diagrams in your answer booklet

(a) ABCD is a parallelogram. AD and QE are produced to meet at R.

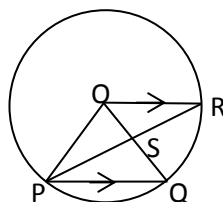
PQ and BE are chords.

(i) Prove that APQR is a cyclic quadrilateral. (6)

(ii) Hence, show that $\angle PCD = \angle DRQ$. (2)



(b) PQ is a chord of the circle with centre O. OR is parallel to PQ. Prove that $\angle RSQ$ is three times the size of $\angle RPQ$. (6)



END OF EXAMINATION ☺

YEAR 11 HSC Ext 1 Task 1 2014.

Q1 B

Q2 C

Q3 B.

Q4 C

Q5 A.

b) $7^{2n} - 3^{3n} \quad (\div 11).$

Show true for $n=1$.

$$7^2 - 3^3$$

$$= 49 - 27$$

$$= 22.$$

$\therefore$ True for $n=1$.

Question 6.

Assume true for $n=k$.

a) $1+4+7+\dots+(3n-2) = \frac{n(3n-1)}{2}$

$$7^{2k} - 3^{3k} = 11P \quad (P \text{ is an integer})$$

$$7^{2k} = 11P + 3^{3k}$$

Show true for $n=1$.

Show true for $n=k+1$.

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1(3-1)}{2}$$

$$7^{2(k+1)} - 3^{3(k+1)} = 11Q$$

$$= 1.$$

(Q is an integer)

$\therefore$ True for $n=1$.

$$\text{LHS} = 7^{2k+2} - 3^{3k+3}$$

$$= 7^2 \times 7^{2k} - 3^3 \times 3^{3k}$$

Assume true for $n=k$.

$$= 49(11P + 3^{3k}) - 27 \times 3^{3k}$$

$$\therefore 1+4+7+\dots+(3k-2) = \frac{k(3k-1)}{2}$$

$$= 49 \times 11P + 49 \times 3^{3k} - 27 \times 3^{3k}$$

Prove true for $n=k+1$

$$= 49 \times 11P + 22 \times 3^{3k}$$

$$\therefore 1+4+7+\dots+(3k-2)+(3k+1) = \frac{(k+1)(3k+2)}{2}$$

$$= 11(49P + 2 \times 3^{3k})$$

$$= 11Q.$$

$\therefore$ If true for $n=k$ then true for $n=k+1$.

$$\text{LHS} = 1+4+7+\dots+(3k-2)+(3k+1)$$

$$= \frac{k(3k-1)}{2} + (3k+1)$$

$\therefore$ By the principle of mathematical induction true for all $n \geq 1$.

$$= \frac{1}{2} [k(3k-1) + 6k+2]$$

$$= \frac{1}{2} (3k^2 - k + 6k + 2)$$

$$= \frac{1}{2} (3k^2 + 5k + 2)$$

$$= \frac{1}{2} (k+1)(3k+2)$$

$$= \text{RHS}.$$

$\therefore$ If true for $n=k$ then true for $n=k+1$.

$\therefore$ By the principle of mathematical induction true for all $n \geq 1$.

Question 6 cont.

c) $n! > 2^n$ $n > 3$.

Show true for $n=4$.

LHS = $4!$ RHS = 2^4

$= 24$ $= 16$.

$\therefore 24 > 16$

$\therefore$ True for $n=4$.

Assume true for $n=k$.

$\therefore k! > 2^k$

Prove true for $n=k+1$.

$\therefore (k+1)! > 2^{k+1}$.

LHS = $(k+1)!$

$= (k+1)k!$

$> (k+1) \times 2^k$

$> 2 \times 2^k$ (as $k > 3$)

$= 2^{k+1}$.

$\therefore (k+1)! > (k+1)2^k > 2^{k+1}$.

$\therefore$ If true for $n=k$ then true for $n=k+1$

$\therefore$ By the principle of mathematical induction true for all $n > 3$.

Question 7

a) $\angle FDC = 90 - 37$

(tangent $\perp$ to radius at point of contact)
 $= 53^\circ$.

$\angle DEF = 53^\circ$

($\angle$ in the alternate segment)

$\angle EGD = 180 - (53 + 90)$

($\angle$ sum of a Δ)

$= 37^\circ$.

b) $\angle EDA = x$

($\angle$ in the alternate segment)

$\angle FDC = x$.

($\angle$ in the alternate segment)

$\angle DCA = x$

($\angle$ in the alternate segment)

$\angle DAC = x$

($\angle$ in the alternate segment)

$\therefore \angle EDA = \angle DAC = x$.

$\therefore$ (alternate $\angle$'s equal)

$\therefore EF \parallel AC$.

c) Let $\angle ADB = x$, $\angle ABE = y$.

$\angle DCE = x$.

($\angle$ in the alternate segment)

$\angle ECB = y$.

($\angle$ in the alternate segment)

$\angle DAB = 180 - (x + y)$

($\angle$ sum of a Δ)

$\therefore \angle DAB + \angle DCB = 180 - (x + y) + (x + y)$
 $= 180$

(supplementary)

$\therefore$ ABCD is a cyclic quadrilateral as opposite angles are supplementary.

Question 8.

$$\begin{aligned} a) T_4 &= {}^8C_3 \left(\frac{m}{2}\right)^5 (3n)^3 \\ &= {}^8C_3 \times \frac{m^5}{32} \times 27n^3 \\ &= \frac{27}{32} \times {}^8C_3 m^5 n^3 \end{aligned}$$

$$b) {}^7C_5 2^2 (-x)^5$$

$$= -84x^5$$

$\therefore$ The coefficient is -84 .

$$c) \left(x - \frac{1}{x}\right)^6$$

$$\begin{aligned} &= {}^6C_0 x^6 - {}^6C_1 x^4 + {}^6C_2 x^2 - {}^6C_3 + {}^6C_4 x^{-2} \\ &\quad - {}^6C_5 x^{-4} + {}^6C_6 x^{-6} \end{aligned}$$

$$\left(x + \frac{1}{x}\right)^8$$

$$\begin{aligned} &= {}^8C_0 x^8 + {}^8C_1 x^6 + {}^8C_2 x^4 + {}^8C_3 x^2 \\ &\quad + {}^8C_4 + {}^8C_5 x^{-2} + {}^8C_6 x^{-4} + {}^8C_7 x^{-6} + {}^8C_8 x^{-8} \end{aligned}$$

$\therefore$ coefficient x^{-8}

$$\begin{aligned} &= \left({}^8C_5 \times {}^6C_6\right) + \left({}^8C_6 \times \left(-{}^6C_5\right)\right) \\ &\quad + \left({}^8C_7 \times {}^6C_4\right) + \left({}^8C_8 \times \left(-{}^6C_3\right)\right) \end{aligned}$$

$$= 56 - 168 + 120 - 20$$

$$= -12$$

$\therefore$ The coefficient is -12 .

Question 8. cont.

$$d) T_4 = {}^nC_3 2^{n-3} (3x)^3$$

$$T_5 = {}^nC_4 2^{n-4} (3x)^4$$

$$\frac{{}^nC_3 2^{n-3} 3^3}{{}^nC_4 2^{n-4} 3^4} = \frac{8}{15}$$

$$\frac{{}^nC_3}{{}^nC_4} \times \frac{2}{3} = \frac{8}{15}$$

nC_4

$$\frac{4}{n-3} \times \frac{2}{3} = \frac{8}{15}$$

$$\therefore \frac{8}{3(n-3)} = \frac{8}{15}$$

$$\therefore 3(n-3) = 15$$

$$n-3 = 5$$

$$\therefore n = 8$$

Question 9

$$a) (5+2x)^{12}$$

$$i) T_{k+1} = {}^{12}C_k 5^{12-k} (2x)^k$$

$$T_k = {}^{12}C_{k-1} 5^{13-k} (2x)^{k-1}$$

$$\begin{aligned} \therefore \frac{T_{k+1}}{T_k} &= \frac{{}^{12}C_k 5^{12-k} 2^k x^k}{{}^{12}C_{k-1} 5^{13-k} 2^{k-1} x^{k-1}} \\ &= \frac{12!}{k! (12-k)!} \times \frac{2}{5} x \\ &= \frac{12!}{(k-1)! (13-k)!} \end{aligned}$$

Question 9 cont.

$$\text{ii) } = \frac{(13+k)}{k} \times \frac{2}{5} x$$

$$= \frac{2(13-k)}{5k} x$$

$$\frac{26-2k}{5k} > 1$$

$$5k < 26-2k$$

$$k < 3.7 \quad \therefore k = 3$$

$\therefore$ The greatest coefficient is.

$$T_4 = {}^{12}C_3 5^9 2^3$$

$$\text{iii) } \frac{2(13-k)}{5k} x > 1$$

$$\text{When } x = \frac{1}{2}$$

$$\frac{2(13-k)}{5k} \cdot \frac{1}{2} > 1$$

$$13-k > 5k$$

$$13 > 6k$$

$$\therefore k < 2.1$$

$$\therefore k = 2$$

$$T_3 = {}^{12}C_2 5^{10} 2^2 \left(\frac{1}{2}\right)^2$$

$$= {}^{12}C_2 5^{10}$$

$$\text{c) } T_2 = {}^{nC_1} x \quad T_3 = {}^{nC_2} x^2$$

$$T_4 = {}^{nC_3} x^3$$

$$T_3 - T_2 = T_4 - T_3$$

$${}^nC_2 - {}^nC_1 = {}^nC_3 - {}^nC_2$$

$$\frac{n!}{2!(n-2)!} - \frac{n!}{(n-1)!} = \frac{n!}{3!(n-3)!} - \frac{n!}{2!(n-2)!}$$

$$\frac{n!(n-3)}{2!(n-1)!} = \frac{n!(n-5)}{3!(n-2)!}$$

$$\therefore 3n!(n-3) = n!(n-1)(n-5)$$

$$3(n-3) = (n-1)(n-5)$$

$$3n-9 = n^2-6n+5$$

$$n^2-9n+14=0$$

$$(n-7)(n-2)=0$$

$$\therefore n=7$$

Question 10.

a) i) Let $\angle PCD = x$.

$$\angle BPC = x$$

(alternate $\angle$'s, $AB \parallel DC$)

Let $\angle PBE = y$.

$$\angle PQE = y.$$

($\angle$'s subtended at circumference by arc PE)

$$\angle BAD = 180 - y$$

(co-interior $\angle$'s, $BC \parallel AD$).

$$\therefore \angle BAD + \angle PQE = 180.$$

(supplementary).

$\therefore$ APQR is a cyclic quadrilateral.

(opposite $\angle$'s are supplementary)

ii) $\angle APC = 180 - x$

(straight $\angle$)

$$\therefore \angle DRQ = x$$

(opposite $\angle$'s of a cyclic quadrilateral)

$$\therefore \angle PCD = \angle DRQ.$$

b) Let $\angle RSQ = y$

and $\angle RPQ = x$.

$$\angle ROQ = 2x.$$

($\angle$ at the centre is twice $\angle$ at the circumference on the same arc)

$$\angle ORS = x$$

(alternate $\angle$'s, $OR \parallel PQ$).

$$\begin{aligned} \therefore \angle RSQ &= 2x + x \\ (\text{exterior } \angle \text{ of a } \Delta) \\ &= 3x \end{aligned}$$

$$\therefore y = 3x.$$